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NormaBRAIN

260728_NormaSummer

AAhead_64ch tfl_b1map_sag mp2rage_wip_sag_1iso_c6DL_5340us mp2rage_wip_sag_1iso_c6cs7DL_6820us vibeMT_mtw_3eco_sag_1iso_c9 vibeMT_mt0_6eco_sag_1iso_c9 vibeMT_t1w_6eco_sag_1iso_c9 vibeMT_pdw_6eco_sag_1iso_c9 ihMT_MC_sag_1.6iso_p3 t2_semc_wip_tra_1x1x1_p5 t2_spc_flair_mt_wip_sag_1iso_c6 t2s_ep3d_wip_sag_0p65_c4DL pcasl_3d_wip_2.5iso_tra_singleTI_DL dwi_3sh_b2k_tra_PA_1p8_p1s3 dwi_3sh_b2k_tra_AP_1p8_p1s3
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TA: 14 sec Coil Selection: Auto Voxel Size: 1.6×1.6×1.6 mm³ Acc:: 3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	On
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	128
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	260 mm
FOV Phase	100.0 %
Slice Thickness	1.6 mm
TR	3.15 ms
TE	1.37 ms
Averages	1
Concatenations	1
AutoAlign	Head

Contrast - Common

TR	3.15 ms
TE	1.37 ms
Flip Angle	8 deg
Fat-Water Contrast	Standard
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Time to Center	6.2 s

Resolution - Common

FOV Read	260 mm
FOV Phase	100.0 %

Resolution - Common

Slice Thickness	1.6 mm
Base Resolution	160
Phase Resolution	100 %
Slice Resolution	69 %
Trajectory	Cartesian

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	3
Reference Lines PE	24
Acceleration Factor 3D	1
Phase Partial Fourier	6/8
Slice Partial Fourier	6/8
Asymmetric Echo	Weak

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	3D
Normalize	Prescan
Noise Masking	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	128
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	260 mm
FOV Phase	100.0 %
Slice Thickness	1.6 mm
TR	3.15 ms
Multi-Slice Mode	Sequential
Series	Ascending
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	Head
Initial Position	Isocenter
L	0.0 mm

Geometry - AutoAlign

P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
CoilShim	Off
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.259673 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Inline - MapIt

MapIt	None
Flip Angle	8 deg
Measurements	1
Contrasts	1
TE	1.37 ms
TR	3.15 ms
Save Original Images	On

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	fl
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Normal
Bandwidth	540 Hz/Px
Asymmetric Echo	Weak

Sequence - Part 2

Introduction	On
RF Spoiling	On
Breast Application	Off
Phase Enc. Order	Automatic

Sequence - Assistant

SAR Assistant	Off
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TA: 1:00 min Coil Selection: Auto Voxel Size: 3.0×3.0×3.0 mm³ Acc:: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
Graphic segment	Default
Inline Movie	Off

Resolution - Common

Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	Off
Asymmetric Echo	Allowed

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	2D
Normalize	Off
Image Filter	Off

Routine

Slice Group	1
Slices	50
Distance Factor	25 %
Position	L2.4 P2.5 F25.5 mm
Orientation	S > C-5.5 > T-0.6
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	288 mm
FOV Phase	93.8 %
Slice Thickness	3.0 mm
TR	30000.0 ms
TE	2.24 ms
Averages	1
Concatenations	1
AutoAlign	Head > Basis

Geometry - Common

Slice Group	1
Slices	50
Distance Factor	25 %
Position	L2.4 P2.5 F25.5 mm
Orientation	S > C-5.5 > T-0.6
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	288 mm
FOV Phase	93.8 %
Slice Thickness	3.0 mm
TR	30000.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Contrast - Common

TR	30000.0 ms
TE	2.24 ms
Magn. Preparation	None
Flip Angle	8 deg
Fat-Water Contrast	Standard
Reconstruction	Magnitude

Geometry - AutoAlign

Slice Group	1
Position	L2.4 P2.5 F25.5 mm
Orientation	S > C-5.5 > T-0.6
Phase Encoding Dir.	A >> P
AutoAlign	Head > Basis
Initial Position	L0.0 A5.6 F26.5
R	0.0 mm
A	5.6 mm
F	26.5 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	288 mm
FOV Phase	93.8 %
Slice Thickness	3.0 mm
Base Resolution	96

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

Sequence - Part 2

Introduction	Off
RF Spoiling	On

Sequence - Assistant

SAR Assistant	Off
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System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
CoilShim	Off
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	L2.6 P25.5 F18.2 mm
! Orientation	S > C-5.8
! Rotation	-6.11 deg
! A >> P	183 mm
! F >> H	264 mm
! R >> L	143 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slice-sel.

System - Tx/Rx

Frequency 1H	123.259673 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	tfl
Dimension	2D
Excitation	Slice-sel.
RF Pulse Type	Low SAR
Gradient Mode	Normal
Flow Compensation	None
Bandwidth	490 Hz/Px
Echo Spacing	4.56 ms
Asymmetric Echo	Allowed
Turbo Factor	90

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TA: 3:30 min Coil Selection: Auto Voxel Size: 1.0x1.0x1.0 mm³ Acc:: 6 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	L1.9 P7.9 F27.2 mm
Orientation	S > C-5.5 > T-0.6
Phase Encoding Dir.	A >> P
Slices per Slab	192
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	288 mm
FOV Phase	86.1 %
Slice Thickness	1.00 mm
TR	5000.0 ms
TE	2.17 ms
Averages	1
Concatenations	1
AutoAlign	Head > Basis

Contrast - Common

TR	5000.0 ms
TE	2.17 ms
Magn. Preparation	Non-sel. IR
TI 1	700 ms
TI 2	2500 ms
Flip Angle 1	4 deg
Flip Angle 2	5 deg
Fat-Water Contrast	Standard
Dark Blood	On
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Contrast - Dynamic

Reordering	Linear
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Resolution - Common

FOV Read	288 mm
FOV Phase	86.1 %
Slice Thickness	1.00 mm
Base Resolution	288
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAPIRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	3
Reference Lines PE	32
Acceleration Factor 3D	2
Reference Lines 3D	32
Reordering Shift 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	3D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	L1.9 P7.9 F27.2 mm
Orientation	S > C-5.5 > T-0.6
Phase Encoding Dir.	A >> P
Slices per Slab	192
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	288 mm
FOV Phase	86.1 %
Slice Thickness	1.00 mm
TR	5000.0 ms
Multi-Slice Mode	Single Shot
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L1.9 P7.9 F27.2 mm
Orientation	S > C-5.5 > T-0.6
Phase Encoding Dir.	A >> P
AutoAlign	Head > Basis
Initial Position	L0.0 A0.6 F29.3
L	0.0 mm
A	0.6 mm
F	29.3 mm
Initial Orientation	Sagittal
Initial Rotation	0.01 deg

Geometry - Navigator**Geometry - Tim Planning Suite**

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	GCC -> 20ch
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
CoilShim	Off
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	L2.6 P25.4 F18.2 mm
! Orientation	S > C-5.8
! Rotation	-6.10 deg
! A >> P	183 mm
! F >> H	264 mm
! R >> L	143 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.259673 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	5000.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	Non-sel. IR
TI 1	700 ms
TI 2	2500 ms
Dark Blood	On
FOV Read	288 mm
FOV Phase	86.1 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Inline - MapIt

MapIt	T1 Map
Flip Angle 1	4 deg
Flip Angle 2	5 deg
Measurements	1
Contrasts	1
TE	2.17 ms
TR	5000.0 ms
Save Original Images	On

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	WIP_tfl
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Fast
Flow Compensation	None
Reordering	Linear
Bandwidth	380 Hz/Px
Echo Spacing	5.34 ms
Asymmetric Echo	Off
Turbo Factor	192

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off

Sequence - Special

Interpolation Mode	Default
DL Recon	DL-CAIPI-P1
Run in Background	Off
Denoising Strength	0.0

Sequence - Assistant

SAR Assistant	Off
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\\USER\ittroalen\NormaBRAIN\260728_NormaSummer\mp2rage_wip_sag_1iso_c6cs7DL_6820us

TA: 3:24 min Coil Selection: Auto Voxel Size: 1.0×1.0×1.0 mm³ Acc:: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	L1.9 P7.9 F27.2 mm
Orientation	S > C-5.5 > T-0.6
Phase Encoding Dir.	A >> P
Slices per Slab	192
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	288 mm
FOV Phase	86.1 %
Slice Thickness	1.00 mm
TR	5000.0 ms
TE	2.88 ms
Averages	1
Concatenations	1
AutoAlign	Head > Basis

Contrast - Common

TR	5000.0 ms
TE	2.88 ms
Magn. Preparation	Non-sel. IR
TI 1	700 ms
TI 2	2500 ms
Flip Angle 1	4 deg
Flip Angle 2	5 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Contrast - Dynamic

Reordering	Linear
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Resolution - Common

FOV Read	288 mm
FOV Phase	86.1 %
Slice Thickness	1.00 mm
Base Resolution	288
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Reference Scans	GRE/Separate
Reference Lines PE	32
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	3D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	L1.9 P7.9 F27.2 mm
Orientation	S > C-5.5 > T-0.6
Phase Encoding Dir.	A >> P
Slices per Slab	192
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	288 mm
FOV Phase	86.1 %
Slice Thickness	1.00 mm
TR	5000.0 ms
Multi-Slice Mode	Single Shot
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L1.9 P7.9 F27.2 mm
Orientation	S > C-5.5 > T-0.6
Phase Encoding Dir.	A >> P

Geometry - AutoAlign

AutoAlign	Head > Basis
Initial Position	L0.0 A0.6 F29.3
L	0.0 mm
A	0.6 mm
F	29.3 mm
Initial Orientation	Sagittal
Initial Rotation	0.01 deg

Geometry - Navigator**Geometry - Tim Planning Suite**

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	GCC -> 20ch
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
CoilShim	Off
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	L2.6 P25.4 F18.2 mm
! Orientation	S > C-5.8
! Rotation	-6.10 deg
! A >> P	183 mm
! F >> H	264 mm
! R >> L	143 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.259673 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off

System - Tx/Rx

Image Scaling	1.000
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Physio - Signal

1st Signal/Mode	None
TR	5000.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	Non-sel. IR
TI 1	700 ms
TI 2	2500 ms
Dark Blood	Off
FOV Read	288 mm
FOV Phase	86.1 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Inline - MapIt

MapIt	T1 Map
Flip Angle 1	4 deg
Flip Angle 2	5 deg
Measurements	1
Contrasts	1
TE	2.88 ms
TR	5000.0 ms
Save Original Images	On

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	WIP_tfl
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Fast
Flow Compensation	None
Reordering	Linear
Bandwidth	240 Hz/Px
Echo Spacing	6.82 ms
Asymmetric Echo	Off
Turbo Factor	170

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off

Sequence - Special

Sparse Undersampling	Poisson Disc
Total Acceleration Factor	7.0 x
Acceleration Factor PE	2
Acceleration Factor 3D	3
Reordering Shift 3D	2
Density Exponent	2.0
Fully Sampled Radius	0.05
Maximum Radius	1.00
Diff. Mask per INV	Off
DL Recon	DL-CS-CAIPI-P1
Denoising Strength	0.0
CS Regularization	0.0010
Uniform Img.	On
Denoised UNI	On
FLAWS-min	Off
FLAWS-hc	Off
FLAWS-hco	Off
Div. Image	Off
UNIDenoised Weight	2.0
Synthetic T11	0 ms
Synthetic T12	0 ms
Synthetic Contrasts	Off

Sequence - Assistant

SAR Assistant	Off
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TA: 2:36 min Coil Selection: Auto Voxel Size: 1.0x1.0x1.0 mm³ Acc:: 9 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Contrast - Dynamic

3D Reordering	Standard
Time to Center	83.2 s
Burn Time to Center	Off

Resolution - Common

FOV Read	288 mm
FOV Phase	86.1 %
Slice Thickness	1.0 mm
Base Resolution	288
Phase Resolution	100 %
Slice Resolution	100 %
Trajectory	Cartesian
Interpolation	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	L1.9 P7.9 F27.2 mm
Orientation	S > C-5.5 > T-0.6
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	288 mm
FOV Phase	86.1 %
Slice Thickness	1.0 mm
TR	30.00 ms
TE 1	2.54 ms
TE 2	7.16 ms
TE 3	11.78 ms
Averages	1
Concatenations	1
AutoAlign	Head > Basis

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
CAIPIRINHA Mode	Free
Total Factor	9
Reference Scans	GRE/Separate
Acceleration Factor PE	3
Reference Lines PE	32
Acceleration Factor 3D	3
Reference Lines 3D	33
Reordering Shift 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Off
Distortion Correction	3D
Normalize	Off
Image Filter	Off

Contrast - Common

TR	30.00 ms
TE 1	2.54 ms
TE 2	7.16 ms
TE 3	11.78 ms
Flip Angle	14 deg
Fat-Water Contrast	Standard
Contrasts	3
Reconstruction	Magn./Phase

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	L1.9 P7.9 F27.2 mm
Orientation	S > C-5.5 > T-0.6
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	288 mm
FOV Phase	86.1 %
Slice Thickness	1.0 mm
TR	30.00 ms

Contrast - Dynamic

Dynamic Mode	Standard
Temporal Interpolation	1
Measurements	1
Multiple Series	Each Measurement

Geometry - Common

Series	Ascending
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L1.9 P7.9 F27.2 mm
Orientation	S > C-5.5 > T-0.6
Phase Encoding Dir.	A >> P
AutoAlign	Head > Basis
Initial Position	L0.0 A0.6 F29.3
L	0.0 mm
A	0.6 mm
F	29.3 mm
Initial Orientation	Sagittal
Initial Rotation	0.01 deg

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
CoilShim	Off
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	L2.6 P25.4 F18.2 mm
! Orientation	S > C-5.8
! Rotation	-6.10 deg
! A >> P	183 mm
! F >> H	264 mm
! R >> L	143 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.259673 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Inline - Composing

Inline Composing	Off
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Inline - MapIt

MapIt	None
Flip Angle	14 deg
Measurements	1
Contrasts	3
TE 1	2.54 ms
TE 2	7.16 ms
TE 3	11.78 ms
TR	30.00 ms

Inline - MapIt

Save Original Images	On
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Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	fl_v3.0
Dimension	3D
Excitation	Non-sel.
Readout Mode	Monopolar
Gradient Mode	Performance
Flow Compensation 1	None
Flow Compensation 2	None
Flow Compensation 3	None
Bandwidth	290 Hz/Px
Optimization	Min. TE

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Incr. Gradient Spoiling	On
Breast Application	Off
Phase Enc. Order	Automatic

Sequence - Special

MT preparation	On
MT Pulse FA	560.0 deg
MT Pulse Frequency	4000.0 Hz
MT Pulse Duration	12.0 ms
MT Grad. Spoiling RO	2.1 cycles/vox
MT Grad. Spoiling SL	2.1 cycles/vox
MT Post-Pulse Delay	2.1 ms
MT Pulse	Hann-Sine
DL Denoising	Off
Buff IcePAT Mem. Alloc.	On
Use SVD Phase Recon.	Off
Enable Implicit Filter	Off
Use SLR Pulse	Off
RO Pulse Duration	250 us
RO Pulse	Rectangular
Total Grad. Spoiling GRE	2.1 cycles/vox
RF Pulse Phase Inc.	50.0 deg
Dummy Scans Duration	4950.0 ms

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s
Optimization	Min. TE

\\USER\l\troalen\NormaBRAIN\260728_NormaSummer\vibeMT_mt0_6eco_sag_1iso_c9

TA: 2:36 min Coil Selection: Auto Voxel Size: 1.0×1.0×1.0 mm³ Acc:: 9 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	L1.9 P7.9 F27.2 mm
Orientation	S > C-5.5 > T-0.6
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	288 mm
FOV Phase	86.1 %
Slice Thickness	1.0 mm
TR	30.00 ms
TE 1	2.54 ms
TE 2	7.16 ms
TE 3	11.78 ms
TE 4	16.40 ms
TE 5	21.02 ms
TE 6	25.64 ms
Averages	1
Concatenations	1
AutoAlign	Head > Basis

Contrast - Common

TR	30.00 ms
TE 1	2.54 ms
TE 2	7.16 ms
TE 3	11.78 ms
TE 4	16.40 ms
TE 5	21.02 ms
TE 6	25.64 ms
Flip Angle	14 deg
Fat-Water Contrast	Standard
Contrasts	6
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
Temporal Interpolation	1
Measurements	1
Multiple Series	Each Measurement
3D Reordering	Standard
Time to Center	83.2 s
Burn Time to Center	Off

Resolution - Common

FOV Read	288 mm
FOV Phase	86.1 %
Slice Thickness	1.0 mm
Base Resolution	288
Phase Resolution	100 %
Slice Resolution	100 %
Trajectory	Cartesian
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
CAIPIRINHA Mode	Free
Total Factor	9
Reference Scans	GRE/Separate
Acceleration Factor PE	3
Reference Lines PE	32
Acceleration Factor 3D	3
Reference Lines 3D	33
Reordering Shift 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Off
Distortion Correction	3D
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	L1.9 P7.9 F27.2 mm
Orientation	S > C-5.5 > T-0.6
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %

Geometry - Common

FOV Read	288 mm
FOV Phase	86.1 %
Slice Thickness	1.0 mm
TR	30.00 ms
Series	Ascending
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L1.9 P7.9 F27.2 mm
Orientation	S > C-5.5 > T-0.6
Phase Encoding Dir.	A >> P
AutoAlign	Head > Basis
Initial Position	L0.0 A0.6 F29.3
L	0.0 mm
A	0.6 mm
F	29.3 mm
Initial Orientation	Sagittal
Initial Rotation	0.01 deg

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
CoilShim	Off
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	L2.6 P25.4 F18.2 mm
! Orientation	S > C-5.8
! Rotation	-6.10 deg

System - Adjust Volume

! A >> P	183 mm
! F >> H	264 mm
! R >> L	143 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.259673 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Inline - Composing

Inline Composing	Off
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Inline - MapIt

MapIt	None
Flip Angle	14 deg

Inline - MapIt

Measurements	1
Contrasts	6
TE 1	2.54 ms
TE 2	7.16 ms
TE 3	11.78 ms
TE 4	16.40 ms
TE 5	21.02 ms
TE 6	25.64 ms
TR	30.00 ms
Save Original Images	On

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	fl_v3.0
Dimension	3D
Excitation	Non-sel.
Readout Mode	Monopolar
Gradient Mode	Performance
Flow Compensation 1	None
Flow Compensation 2	None
Flow Compensation 3	None
Flow Compensation 4	None
Flow Compensation 5	None
Flow Compensation 6	None
Bandwidth	290 Hz/Px
Optimization	Min. TE

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Incr. Gradient Spoiling	On
Breast Application	Off
Phase Enc. Order	Automatic

Sequence - Special

MT preparation	Off
DL Denoising	Off
Buff IcePAT Mem. Alloc.	On
Use SVD Phase Recon.	Off
Enable Implicit Filter	Off
Use SLR Pulse	Off
RO Pulse Duration	250 us
RO Pulse	Rectangular
Total Grad. Spoiling GRE	2.1 cycles/vox
RF Pulse Phase Inc.	50.0 deg
Dummy Scans Duration	4950.0 ms

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s
Optimization	Min. TE

\\USER\ittoalen\NormaBRAIN\260728_NormaSummer\vibeMT_t1w_6eco_sag_1iso_c9

TA: 2:36 min Coil Selection: Auto Voxel Size: 1.0×1.0×1.0 mm³ Acc:: 9 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	L1.9 P7.9 F27.2 mm
Orientation	S > C-5.5 > T-0.6
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	288 mm
FOV Phase	86.1 %
Slice Thickness	1.0 mm
TR	30.00 ms
TE 1	2.54 ms
TE 2	7.16 ms
TE 3	11.78 ms
TE 4	16.40 ms
TE 5	21.02 ms
TE 6	25.64 ms
Averages	1
Concatenations	1
AutoAlign	Head > Basis

Contrast - Common

TR	30.00 ms
TE 1	2.54 ms
TE 2	7.16 ms
TE 3	11.78 ms
TE 4	16.40 ms
TE 5	21.02 ms
TE 6	25.64 ms
Flip Angle	33 deg
Fat-Water Contrast	Standard
Contrasts	6
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
Temporal Interpolation	1
Measurements	1
Multiple Series	Each Measurement
3D Reordering	Standard
Time to Center	83.2 s
Burn Time to Center	Off

Resolution - Common

FOV Read	288 mm
FOV Phase	86.1 %
Slice Thickness	1.0 mm
Base Resolution	288
Phase Resolution	100 %
Slice Resolution	100 %
Trajectory	Cartesian
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
CAIPIRINHA Mode	Free
Total Factor	9
Reference Scans	GRE/Separate
Acceleration Factor PE	3
Reference Lines PE	32
Acceleration Factor 3D	3
Reference Lines 3D	33
Reordering Shift 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Off
Distortion Correction	3D
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	L1.9 P7.9 F27.2 mm
Orientation	S > C-5.5 > T-0.6
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %

Geometry - Common

FOV Read	288 mm
FOV Phase	86.1 %
Slice Thickness	1.0 mm
TR	30.00 ms
Series	Ascending
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L1.9 P7.9 F27.2 mm
Orientation	S > C-5.5 > T-0.6
Phase Encoding Dir.	A >> P
AutoAlign	Head > Basis
Initial Position	L0.0 A0.6 F29.3
L	0.0 mm
A	0.6 mm
F	29.3 mm
Initial Orientation	Sagittal
Initial Rotation	0.01 deg

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
CoilShim	Off
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	L2.6 P25.4 F18.2 mm
! Orientation	S > C-5.8
! Rotation	-6.10 deg

System - Adjust Volume

! A >> P	183 mm
! F >> H	264 mm
! R >> L	143 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.259673 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Inline - Composing

Inline Composing	Off
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Inline - MapIt

MapIt	None
Flip Angle	33 deg

Inline - MapIt

Measurements	1
Contrasts	6
TE 1	2.54 ms
TE 2	7.16 ms
TE 3	11.78 ms
TE 4	16.40 ms
TE 5	21.02 ms
TE 6	25.64 ms
TR	30.00 ms
Save Original Images	On

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	fl_v3.0
Dimension	3D
Excitation	Non-sel.
Readout Mode	Monopolar
Gradient Mode	Performance
Flow Compensation 1	None
Flow Compensation 2	None
Flow Compensation 3	None
Flow Compensation 4	None
Flow Compensation 5	None
Flow Compensation 6	None
Bandwidth	290 Hz/Px
Optimization	Min. TE

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Incr. Gradient Spoiling	On
Breast Application	Off
Phase Enc. Order	Automatic

Sequence - Special

MT preparation	Off
DL Denoising	Off
Buff IcePAT Mem. Alloc.	On
Use SVD Phase Recon.	Off
Enable Implicit Filter	Off
Use SLR Pulse	Off
RO Pulse Duration	250 us
RO Pulse	Rectangular
Total Grad. Spoiling GRE	2.1 cycles/vox
RF Pulse Phase Inc.	50.0 deg
Dummy Scans Duration	4950.0 ms

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s
Optimization	Min. TE

\\USER\ittoalen\NormaBRAIN\260728_NormaSummer\vibeMT_pdw_6eco_sag_1iso_c9

TA: 2:36 min Coil Selection: Auto Voxel Size: 1.0×1.0×1.0 mm³ Acc:: 9 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	L1.9 P7.9 F27.2 mm
Orientation	S > C-5.5 > T-0.6
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	288 mm
FOV Phase	86.1 %
Slice Thickness	1.0 mm
TR	30.00 ms
TE 1	2.54 ms
TE 2	7.16 ms
TE 3	11.78 ms
TE 4	16.40 ms
TE 5	21.02 ms
TE 6	25.64 ms
Averages	1
Concatenations	1
AutoAlign	Head > Basis

Contrast - Common

TR	30.00 ms
TE 1	2.54 ms
TE 2	7.16 ms
TE 3	11.78 ms
TE 4	16.40 ms
TE 5	21.02 ms
TE 6	25.64 ms
Flip Angle	6 deg
Fat-Water Contrast	Standard
Contrasts	6
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
Temporal Interpolation	1
Measurements	1
Multiple Series	Each Measurement
3D Reordering	Standard
Time to Center	83.2 s
Burn Time to Center	Off

Resolution - Common

FOV Read	288 mm
FOV Phase	86.1 %
Slice Thickness	1.0 mm
Base Resolution	288
Phase Resolution	100 %
Slice Resolution	100 %
Trajectory	Cartesian
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
CAIPIRINHA Mode	Free
Total Factor	9
Reference Scans	GRE/Separate
Acceleration Factor PE	3
Reference Lines PE	32
Acceleration Factor 3D	3
Reference Lines 3D	33
Reordering Shift 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Off
Distortion Correction	3D
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	L1.9 P7.9 F27.2 mm
Orientation	S > C-5.5 > T-0.6
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %

Geometry - Common

FOV Read	288 mm
FOV Phase	86.1 %
Slice Thickness	1.0 mm
TR	30.00 ms
Series	Ascending
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L1.9 P7.9 F27.2 mm
Orientation	S > C-5.5 > T-0.6
Phase Encoding Dir.	A >> P
AutoAlign	Head > Basis
Initial Position	L0.0 A0.6 F29.3
L	0.0 mm
A	0.6 mm
F	29.3 mm
Initial Orientation	Sagittal
Initial Rotation	0.01 deg

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
CoilShim	Off
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	L2.6 P25.4 F18.2 mm
! Orientation	S > C-5.8
! Rotation	-6.10 deg

System - Adjust Volume

! A >> P	183 mm
! F >> H	264 mm
! R >> L	143 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.259673 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Inline - Composing

Inline Composing	Off
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Inline - MapIt

MapIt	None
Flip Angle	6 deg

Inline - MapIt

Measurements	1
Contrasts	6
TE 1	2.54 ms
TE 2	7.16 ms
TE 3	11.78 ms
TE 4	16.40 ms
TE 5	21.02 ms
TE 6	25.64 ms
TR	30.00 ms
Save Original Images	On

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	fl_v3.0
Dimension	3D
Excitation	Non-sel.
Readout Mode	Monopolar
Gradient Mode	Performance
Flow Compensation 1	None
Flow Compensation 2	None
Flow Compensation 3	None
Flow Compensation 4	None
Flow Compensation 5	None
Flow Compensation 6	None
Bandwidth	290 Hz/Px
Optimization	Min. TE

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Incr. Gradient Spoiling	On
Breast Application	Off
Phase Enc. Order	Automatic

Sequence - Special

MT preparation	Off
DL Denoising	Off
Buff IcePAT Mem. Alloc.	On
Use SVD Phase Recon.	Off
Enable Implicit Filter	Off
Use SLR Pulse	Off
RO Pulse Duration	250 us
RO Pulse	Rectangular
Total Grad. Spoiling GRE	2.1 cycles/vox
RF Pulse Phase Inc.	50.0 deg
Dummy Scans Duration	4950.0 ms

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s
Optimization	Min. TE

\\USER\l\Troalen\NormaBRAIN\260728_NormaSummer\ihMT_MC_sag_1.6iso_p3

TA: 10:37 min Coil Selection: Auto Voxel Size: 1.6×1.6×1.6 mm³ Acc:: 3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Contrast - Dynamic

Pause after Meas. 6	0.0 s
Multiple Series	Off
Reordering	Free 0

Resolution - Common

FOV Read	288 mm
FOV Phase	88.6 %
Slice Thickness	1.60 mm
Base Resolution	176
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	3
Reference Lines PE	32
Acceleration Factor 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	3D
Normalize	Off
Image Filter	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	L1.9 P7.9 F27.2 mm
Orientation	S > C-5.5 > T-0.6
Phase Encoding Dir.	A >> P
Slices per Slab	112
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	288 mm
FOV Phase	88.6 %
Slice Thickness	1.60 mm
TR	1260.0 ms
TE	2.35 ms
Averages	1
Concatenations	1
AutoAlign	Head > Basis

Contrast - Common

TR	1260.0 ms
TE	2.35 ms
Flip Angle	5 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	7
Pause after Meas. 1	0.0 s
Pause after Meas. 2	0.0 s
Pause after Meas. 3	0.0 s
Pause after Meas. 4	0.0 s
Pause after Meas. 5	0.0 s

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	L1.9 P7.9 F27.2 mm
Orientation	S > C-5.5 > T-0.6
Phase Encoding Dir.	A >> P
Slices per Slab	112
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	288 mm
FOV Phase	88.6 %
Slice Thickness	1.60 mm
TR	1260.0 ms
Multi-Slice Mode	Sequential
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L1.9 P7.9 F27.2 mm
Orientation	S > C-5.5 > T-0.6
Phase Encoding Dir.	A >> P
AutoAlign	Head > Basis
Initial Position	L0.0 A0.6 F29.3
L	0.0 mm
A	0.6 mm
F	29.3 mm
Initial Orientation	Sagittal
Initial Rotation	0.01 deg

Geometry - Navigator**Geometry - Tim Planning Suite**

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
CoilShim	Off
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	L2.6 P25.4 F18.2 mm
! Orientation	S > C-5.8
! Rotation	-6.10 deg
! A >> P	183 mm
! F >> H	264 mm
! R >> L	143 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.259673 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	2.000

Physio - Signal

1st Signal/Mode	None
TR	1260.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Dark Blood	Off
FOV Read	288 mm
FOV Phase	88.6 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	7
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Inline - MapIt

MapIt	None
Flip Angle	5 deg
Measurements	7
Contrasts	1
TE	2.35 ms
TR	1260.0 ms
Save Original Images	On

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	ihMTtfl
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Normal
Gradient Mode	Performance
Flow Compensation	None
Reordering	Free 0
Bandwidth	300 Hz/Px
Echo Spacing	6.46 ms
Asymmetric Echo	Off
Turbo Factor	112

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	On

Sequence - Special

ihMT preparation	On
ihMT acq. scheme	Regular
ihMT acq. type	Both contrasts
ihMT pulse duration	1000 us
ihMT dt	1500 us
Offset frequency	7000 Hz
Flip angle	203 deg
Tukey shape param.	30 /100
# pulses per burst	6
# burst	6
Burst TR	100000 us
ihMT Prep. time	510000 us
ihMT pulse spoiler	7.5 ms.mT/m
ihMT burst spoiler	0.0 ms.mT/m
Relative RO dephasing moment	3.8
ihMT pulse phase cycling	50.0 deg
ihMT partial k-space	100 /100
dicom 16-bit	On
DL Denoising	Off
Dummy TR	0
Dummy Echoes	3
Geometric Coil Compression	0
Echo spacing	6460 us

Sequence - Assistant

SAR Assistant	Off
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\\USER\Ittroalen\NormaBRAIN\260728_NormaSummer\2_semc_wip_tra_1x1x1_p5

TA: 9:14 min Coil Selection: Manual Voxel Size: 1.0x1.0x1.0 mm³ Acc:: 2 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	145
Distance Factor	0 %
Position	L1.9 P6.8 H33.1 mm
Orientation	T > C3.1 > S0.3
Phase Encoding Dir.	R >> L
Phase Oversampling	0 %
FOV Read	224 mm
FOV Phase	93.8 %
Slice Thickness	1.0 mm
TR	7680.0 ms
TE 1	12.50 ms
TE 2	25.00 ms
TE 3	37.50 ms
TE 4	50.00 ms
TE 5	62.50 ms
TE 6	75.00 ms
TE 7	87.50 ms
TE 8	100.00 ms
TE 9	112.50 ms
TE 10	125.00 ms
TE 11	137.50 ms
TE 12	150.00 ms
TE 13	162.50 ms
TE 14	175.00 ms
TE 15	187.50 ms
TE 16	200.00 ms
Averages	1
Concatenations	4
AutoAlign	Head > Brain
Coil Elements	HC1-6

Contrast - Common

TR	7680.0 ms
TE 1	12.50 ms

Contrast - Common

TE 2	25.00 ms
TE 3	37.50 ms
TE 4	50.00 ms
TE 5	62.50 ms
TE 6	75.00 ms
TE 7	87.50 ms
TE 8	100.00 ms
TE 9	112.50 ms
TE 10	125.00 ms
TE 11	137.50 ms
TE 12	150.00 ms
TE 13	162.50 ms
TE 14	175.00 ms
TE 15	187.50 ms
TE 16	200.00 ms
TD	0.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	180 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	16
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	224 mm
FOV Phase	93.8 %
Slice Thickness	1.0 mm
Base Resolution	224
Phase Resolution	80 %

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	2

Resolution - Filter

Raw Filter	Off
Elliptical Filter	On
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slice Group	1
-------------	---

Geometry - Common

Slices	145
Distance Factor	0 %
Position	L1.9 P6.8 H33.1 mm
Orientation	T > C3.1 > S0.3
Phase Encoding Dir.	R >> L
Phase Oversampling	0 %
FOV Read	224 mm
FOV Phase	93.8 %
Slice Thickness	1.0 mm
TR	7680.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	4

Geometry - AutoAlign

Slice Group	1
Position	L1.9 P6.8 H33.1 mm
Orientation	T > C3.1 > S0.3
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	L0.1 A8.2 F0.6
L	0.1 mm
A	8.2 mm
F	0.6 mm
Initial Orientation	Transversal
Initial Rotation	90.01 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Matrix Optimization	Performance

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Advanced
B1 Shim	TrueForm
CoilShim	Off
Adjustment Tolerance	Auto
Adjust with Body Coil	Off

System - Adjustments

Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	L1.0 P15.0 H34.1 mm
! Orientation	T > C3.1 > S0.3
! Rotation	84.43 deg
! R >> L	162 mm
! A >> P	201 mm
! F >> H	144 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
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System - Tx/Rx

Frequency 1H	123.259673 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	7680.0 ms
Concatenations	4

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	224 mm
FOV Phase	93.8 %
Phase Resolution	80 %

Physio - PACE

Resp. Control	Off
Concatenations	4

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off

Inline - MIP

Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Sequence - Special

Simulated TE 1	0 ms
Simulated TE 2	0 ms
Simulated TE 3	0 ms
Simulated TE 4	0 ms

Inline - Composing

Inline Composing	Off
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Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

Inline - MapIt

MapIt	None
Flip Angle	180 deg
Measurements	1
Contrasts	16
TE 1	12.50 ms
TE 2	25.00 ms
TE 3	37.50 ms
TE 4	50.00 ms
TE 5	62.50 ms
TE 6	75.00 ms
TE 7	87.50 ms
TE 8	100.00 ms
TE 9	112.50 ms
TE 10	125.00 ms
TE 11	137.50 ms
TE 12	150.00 ms
TE 13	162.50 ms
TE 14	175.00 ms
TE 15	187.50 ms
TE 16	200.00 ms
TR	7680.0 ms
Save Original Images	On

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	WIP_se
RF Pulse Type	Low SAR
Gradient Mode	Fast
Bandwidth	221 Hz/Px

Sequence - Part 2

Introduction	On
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Sequence - Special

Undersampling Factor	5
R2 Images	On
M0 Images	On
Interpolate	On
Filter Background	On
Regularization:	5.00
Virtual Channels	4
Simulated TE 0	80 ms

\\USER\Ittroalen\NormaBRAIN\260728_NormaSummer\T2_spc_flair_mt_wip_sag_1iso_c6

TA: 3:57 min Coil Selection: Auto Voxel Size: 0.5×0.5×0.5 mm³ Acc:: 6 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Position	L1.9 P7.9 F27.2 mm
Orientation	S > C-5.5 > T-0.6
Phase Encoding Dir.	A >> P
Slices per Slab	352
Phase Oversampling	10 %
Slice Oversampling	0.0 %
FOV Read	288 mm
FOV Phase	86.1 %
Slice Thickness	0.50 mm
TR	7000.0 ms
TE	371.00 ms
Averages	1.0
Concatenations	1
AutoAlign	Head > Basis

Contrast - Common

TR	7000.0 ms
TE	371.00 ms
Magn. Preparation	Non-sel. T2 prep. IR
TI	2050 ms
T2 Prep. Duration	125 ms
Flip Angle Mode	Variable
Start flip angle	120 deg
Minimum flip angle	20 deg
Center flip angle	70 deg
Last flip angle	120 deg
Fat-Water Contrast	Fat Saturation
Fat Saturation	Strong
Dark Blood	Off
Blood Suppression	Off
Excitation offset frequency	0 Hz
Wrap-up Magn.	None
Refoc. offset frequency	0 Hz

Contrast - Common

Reconstruction	Magnitude
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Contrast - Dynamic

goKART	Off
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Contrast - ASL

MTC	On
MTC frequency sign	Positive
MTC # RF	10
MTC pulse duration	7680.0 μ s
MTC frequency offset	1200 Hz
MTC pause between RF	0 ms

Resolution - Common

FOV Read	288 mm
FOV Phase	86.1 %
Slice Thickness	0.50 mm
Base Resolution	288
Phase Resolution	90 %
Slice Resolution	50 %
Interpolation	On

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Total Factor	6
Reference Scans	GRE/Separate
Acceleration Factor PE	3
Reference Lines PE	42
Acceleration Factor 3D	2
Reference Lines 3D	42
Reordering Shift 3D	1
Deep Resolve	Boost
Phase Partial Fourier	Allowed
Slice Partial Fourier	7/8
DRS-Interpolation Mode	3xD-PX-HDC-CPU
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	On
Elliptical Filter	Off
Disable Inline Image filter	Off
Distortion Correction	3D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
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Geometry - Common

Slabs	1
Position	L1.9 P7.9 F27.2 mm
Orientation	S > C-5.5 > T-0.6
Phase Encoding Dir.	A >> P
Slices per Slab	352
Phase Oversampling	10 %
Slice Oversampling	0.0 %
FOV Read	288 mm
FOV Phase	86.1 %
Slice Thickness	0.50 mm
TR	7000.0 ms
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L1.9 P7.9 F27.2 mm
Orientation	S > C-5.5 > T-0.6
Phase Encoding Dir.	A >> P
AutoAlign	Head > Basis
Initial Position	L0.0 A0.6 F29.3
L	0.0 mm
A	0.6 mm
F	29.3 mm
Initial Orientation	Sagittal
Initial Rotation	0.01 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
CoilShim	Auto
Adjustment Tolerance	None

System - Adjustments

Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off
Adj Freq. protocol related	On

System - Adjust Volume

! Position	L2.6 P25.4 F18.2 mm
! Orientation	S > C-5.8
! Rotation	-6.10 deg
! A >> P	183 mm
! F >> H	264 mm
! R >> L	143 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.259673 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000
Gain	High

Physio - Signal

1st Signal/Mode	None
Trigger Delay	0 ms
TR	7000.0 ms
Rd crusher scale	1800
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Fat Saturation
Magn. Preparation	Non-sel. T2 prep. IR
TI	2050 ms
T2 Prep. Duration	125 ms
Dark Blood	Off
FOV Read	288 mm
FOV Phase	86.1 %
Phase Resolution	90 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
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Inline - MIP

MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

Inline - Composing

Inline Composing	Off
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Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	wip-spc-t2p+ir
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	None
Reordering	Linear
Bandwidth	599 Hz/Px
Echo Spacing	4.22 ms
Concomitant field corr. R	Product
Turbo Factor	200
Echo Train Duration	789 ms
Rd crusher scale	1800

Sequence - Part 2

Introduction	On
Rel. crusher mom. P	0 %
Rel crusher mom. S	0 %

Sequence - Special

IR refoc pulse	IR10240H180.IR180_36B1_1
IR RF duration	10240 us
IR HS mu	5.00
IR HS BWTP	10.99
IR flip angle	360 deg
T2-Prep. excit/tip-up pulse	RECT
T2-Prep. refoc pulse	IR10240H180.IR180_36B1_1
T2-Prep. # refoc. RF	4
T2-Prep. excit. duration	800 us
T2-Prep. refoc. duration	10240 us
T2-Prep. spoil mom. 10 ³ x	-71.20 usmT/m
T2-Prep. B1 scale	1.00
T2-Prep. crusher mom. rel.	0.00
T2-Prep. Refoc shift	0.00 %
Excit. flip angle	90 deg
Blood Suppression	Off
Trajectory Optimization	8

\\USER\Ittroalen\NormaBRAIN\260728_NormaSummerlt2s_ep3d_wip_sag_0p65_c4DL

TA: 2:34 min Coil Selection: Manual Voxel Size: 0.7×0.7×0.7 mm³ Acc:: 4 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	-50 %
Position	L0.8 P19.5 F33.5 mm
Orientation	S > C-5.5 > T-0.6
Phase Encoding Dir.	A >> P
Slices per Slab	256
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	288 mm
FOV Phase	72.2 %
Slice Thickness	0.65 mm
TR	65.0 ms
TE	32.00 ms
Averages	1
Concatenations	1
AutoAlign	Head > Basis
Coil Elements	HC1-7;NC1,2

Contrast - Common

TR	65.0 ms
TE	32.00 ms
MTC	Off
Flip Angle	10 deg
Fat-Water Contrast	Fast Water Excitation
SWI	On
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Off
3D Reordering	Standard
Reordering	Linear

Resolution - Common

FOV Read	288 mm
FOV Phase	72.2 %
Slice Thickness	0.65 mm
Base Resolution	432
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
CAIPIRINHA Mode	Free
Reference Scans	GRE/Separate
Acceleration Factor PE	2
Reference Lines PE	24
Acceleration Factor 3D	2
Reference Lines 3D	24
Reordering Shift 3D	1
Deep Resolve	Boost
Phase Partial Fourier	Off
Slice Partial Fourier	Off

Resolution - Filter

Raw Filter	On
Elliptical Filter	Off
Hamming	Off
Distortion Correction	3D
Static Field Correction	Off
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	-50 %
Position	L0.8 P19.5 F33.5 mm
Orientation	S > C-5.5 > T-0.6
Phase Encoding Dir.	A >> P
Slices per Slab	256
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	288 mm
FOV Phase	72.2 %
Slice Thickness	0.65 mm
TR	65.0 ms
Multi-Slice Mode	Sequential
Series	Ascending
Concatenations	1

Geometry - AutoAlign

Slab Group	1
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Geometry - AutoAlign

Position	L0.8 P19.5 F33.5 mm
Orientation	S > C-5.5 > T-0.6
Phase Encoding Dir.	A >> P
AutoAlign	Head > Basis
Initial Position	L0.0 P9.5 F37.9
L	0.0 mm
P	9.5 mm
F	37.9 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	GCC -> 12ch

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
CoilShim	Off
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	L2.6 P25.4 F18.2 mm
! Orientation	S > C-5.8
! Rotation	-6.10 deg
! A >> P	183 mm
! F >> H	264 mm
! R >> L	143 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	123.259673 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	65.0 ms
Segments	9
Concatenations	1

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	On
Temp. Highpass Filter	On
Threshold	4.00
Paradigm Size	3
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Active
Spatial Filter	Off
Measurements	1
Multiple Series	Off

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	WIP_EPsw_r
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Fast*
Flow Compensation	On
Reordering	Linear
Bandwidth	304 Hz/Px
Echo Spacing	3.61 ms
Free Echo Spacing	Off
Segments	9
EPI Factor	9

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Phase Correction	External

Sequence - Special

Prep/dummy scan duration	2 sec
Water Excitation options	Mono-polar 1-2-1
Desired RF TBW	4.0
Flyback	No

Sequence - Special

QuickSat	1
Excitation Orientation	SS - Slice Selective
Ernst Angle T1	850 ms
Reduce Slab Thickness?	None
skipped-CAIPI?	Off
Log Signals	Off
Phase Unwrap	Off

Sequence - Assistant

SAR Assistant	Off
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\\USER\ltoalen\NormaBRAIN\260728_NormaSummer\pcasl_3d_wip_2.5iso_tra_singleTI_DL

TA: 5:48 min Coil Selection: Auto Voxel Size: 1.3×1.3×2.5 mm³ Acc:: 4 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Contrast - ASL

Suppression	Gray-White-Strong
Labeling Duration	1800 ms
Postlabeling Delay	1800 ms
Delay Array Size	1

Resolution - Common

FOV Read	240 mm
FOV Phase	100.0 %
Slice Thickness	2.5 mm
Base Resolution	96
Phase Resolution	100 %
Interpolation	On

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
CAIPIRINHA Mode	Free
Reference Scans	GRE/Separate
Acceleration Factor PE	2
Reference Lines PE	32
Acceleration Factor 3D	2
Reference Lines 3D	32
Reordering Shift 3D	1
Deep Resolve	Boost
Phase Partial Fourier	7/8
Slice Partial Fourier	Off
DRS-Deep Resolve Sharp	3xD-PX-HDC-CPU

Routine

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	L1.1 P14.8 H32.8 mm
Orientation	T > C-3.6 > S1.0
Phase Encoding Dir.	P >> A
Slices per Slab	60
Phase Oversampling	0 %
Slice Oversampling	7.5 %
FOV Read	240 mm
FOV Phase	100.0 %
Slice Thickness	2.5 mm
TR	4000.0 ms
TE	17.70 ms
Averages	1
Concatenations	1
AutoAlign	Head > Brain

Resolution - Filter

Raw Filter	On
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Static Field Correction	Off
Normalize	Prescan

Contrast - Common

TR	4000.0 ms
TE	17.70 ms
Flip Angle	160 deg
Fat-Water Contrast	Fat Saturation
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	10
Multiple Series	Off
Delay in TR	0.00 ms
Reordering	Centric

Contrast - ASL

Perfusion Mode	PCASL
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Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	L1.1 P14.8 H32.8 mm
Orientation	T > C-3.6 > S1.0
Phase Encoding Dir.	P >> A
Slices per Slab	60
Phase Oversampling	0 %
Slice Oversampling	7.5 %
FOV Read	240 mm
FOV Phase	100.0 %
Slice Thickness	2.5 mm
TR	4000.0 ms
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L1.1 P14.8 H32.8 mm
Orientation	T > C-3.6 > S1.0
Phase Encoding Dir.	P >> A
AutoAlign	Head > Brain
Initial Position	L0.0 A0.2 F1.3
R	0.0 mm
A	0.2 mm
F	1.3 mm
Initial Orientation	T > C
T > C	-6.80
> S	0.00
Initial Rotation	179.23 deg

Geometry - Saturation

Special Saturation	Parallel F
Gap	60.00 mm

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
CoilShim	Off
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L1.1 P14.8 H32.8 mm
Orientation	T > C-3.6 > S1.0
Rotation	173.74 deg
A >> P	240 mm
R >> L	240 mm
F >> H	150 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
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System - Tx/Rx

Frequency 1H	123.259673 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	4000.0 ms
Segments	4
Concatenations	1

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	WIPtgse
Dimension	3D
RF Pulse Type	Normal
Gradient Mode	Fast
Reordering	Centric
Bandwidth	2480 Hz/Px
Echo Spacing	0.61 ms
DRS-Deep Resolve Sharp	3xD-PX-HDC-CPU
Turbo Factor	16
Segments	4
EPI Factor	21

Sequence - Part 2

Introduction	Off
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Sequence - Special

Masking	Off
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Sequence - Assistant

SAR Assistant	Off
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\\USER\ittoalen\NormaBRAIN\260728_NormaSummer\ldwi_3sh_b2k_tra_PA_1p8_p1s3

TA: 6:27 min Coil Selection: Auto Voxel Size: 1.8×1.8×1.8 mm³ Acc:: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	On
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	81
Distance Factor	0 %
Position	L0.1 A8.2 F0.6 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	212 mm
FOV Phase	100.0 %
Slice Thickness	1.8 mm
TR	3340.0 ms
TE	83.00 ms
Multi-band accel. factor	3
AutoAlign	Head > Brain

Contrast - Common

TR	3340.0 ms
TE	83.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	78 deg
Refocus flip angle	160 deg
Fat-Water Contrast	Standard
Grad. rev. fat suppr.	Enabled
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
Multiple Series	Off
Delay in TR	0.00 ms

Resolution - Common

FOV Read	212 mm
FOV Phase	100.0 %

Resolution - Common

Slice Thickness	1.8 mm
Base Resolution	118
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	6/8

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Static Field Correction	Off
Normalize	Prescan
Noise Masking	Off

Geometry - Common

Slice Group	1
Slices	81
Distance Factor	0 %
Position	L0.1 A8.2 F0.6 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	212 mm
FOV Phase	100.0 %
Slice Thickness	1.8 mm
TR	3340.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Multi-band accel. factor	3

Geometry - AutoAlign

Slice Group	1
Position	L0.1 A8.2 F0.6 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.1 A8.2 F0.6
L	0.1 mm
A	8.2 mm
F	0.6 mm
Initial Orientation	Transversal
Initial Rotation	0.02 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	1 mm
Table Position	F
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Performance
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Advanced
B1 Shim	TrueForm
CoilShim	Off
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	L0.1 A2.3 F0.4 mm
! Orientation	Transversal
! Rotation	0.02 deg
! A >> P	181 mm
! R >> L	141 mm
! F >> H	146 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.259673 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3340.0 ms
Multi-band accel. factor	3

Physio - PACE

Resp. Control	Off
Multi-band accel. factor	3

Diff

Diffusion Mode	Free
Diff. Directions	108
Diffusion Scheme	Monopolar
Diff. Weightings	2
b-value 1	0 s/mm ²
b-value 2	2000 s/mm ²
Averages 1	1
Averages 2	1
Dynamic Field Correction	Off
Invert Gray Scale	Off
Diff. Weighted Images	On
Trace Weighted Images	Off
Tensor	On
FA Maps	Off
ADC Maps	Off
Exponential ADC Maps	Off
Noise Masking	Off
Calculated Image	Off

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	epse
Dimension	2D
Excitation	Standard
Gradient Mode	Performance
Bandwidth	1842 Hz/Px
Echo Spacing	0.65 ms
Free Echo Spacing	Off
EPI Factor	118

Sequence - Part 2

Introduction	Off
RF Spoiling	Off
Phase Correction	Internal

Sequence - Special

Excite pulse duration	3200 us
Refocus pulse duration	7040 us
Min. prep scans	0
Min. prep scans SB	1
Delay before PC scans	0 us
Single-band images	Off
MB LeakBlock kernel	On
MB dual kernel	Off
MB RF phase scramble	Off
Opt. MB RF pulse BW	Off
Time-shifted MB RF	Off
SENSE1 coil combine	On
Invert RO/PE polarity	On
Force Maxwell corr.	Off
PF omits higher k-space	Off
Disable B1 control loop	Off

Sequence - Special

Disable freq. update	Off
Suppress 16-bit DICOM	Off
Force equal slice timing	Off
Online multi-band recon.	Online
FFT scale factor	1.00
Physio recording	Off

Sequence - Assistant

SAR Assistant	Off
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\\USER\l\troalen\NormaBRAIN\260728_NormaSummer\ldwi_3sh_b2k_tra_AP_1p8_p1s3

TA: 6:27 min Coil Selection: Auto Voxel Size: 1.8×1.8×1.8 mm³ Acc:: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	On
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	81
Distance Factor	0 %
Position	L0.1 A8.2 F0.6 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	212 mm
FOV Phase	100.0 %
Slice Thickness	1.8 mm
TR	3340.0 ms
TE	83.00 ms
Multi-band accel. factor	3
AutoAlign	Head > Brain

Contrast - Common

TR	3340.0 ms
TE	83.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	78 deg
Refocus flip angle	160 deg
Fat-Water Contrast	Standard
Grad. rev. fat suppr.	Enabled
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
Multiple Series	Off
Delay in TR	0.00 ms

Resolution - Common

FOV Read	212 mm
FOV Phase	100.0 %

Resolution - Common

Slice Thickness	1.8 mm
Base Resolution	118
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	6/8

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Static Field Correction	Off
Normalize	Prescan
Noise Masking	Off

Geometry - Common

Slice Group	1
Slices	81
Distance Factor	0 %
Position	L0.1 A8.2 F0.6 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	212 mm
FOV Phase	100.0 %
Slice Thickness	1.8 mm
TR	3340.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Multi-band accel. factor	3

Geometry - AutoAlign

Slice Group	1
Position	L0.1 A8.2 F0.6 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.1 A8.2 F0.6
L	0.1 mm
A	8.2 mm
F	0.6 mm
Initial Orientation	Transversal
Initial Rotation	0.02 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	1 mm
Table Position	F
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Performance
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Advanced
B1 Shim	TrueForm
CoilShim	Off
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	L0.1 A2.3 F0.4 mm
! Orientation	Transversal
! Rotation	0.02 deg
! A >> P	181 mm
! R >> L	141 mm
! F >> H	146 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.259673 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3340.0 ms
Multi-band accel. factor	3

Physio - PACE

Resp. Control	Off
Multi-band accel. factor	3

Diff

Diffusion Mode	Free
Diff. Directions	108
Diffusion Scheme	Monopolar
Diff. Weightings	2
b-value 1	0 s/mm ²
b-value 2	2000 s/mm ²
Averages 1	1
Averages 2	1
Dynamic Field Correction	Off
Invert Gray Scale	Off
Diff. Weighted Images	On
Trace Weighted Images	Off
Tensor	On
FA Maps	Off
ADC Maps	Off
Exponential ADC Maps	Off
Noise Masking	Off
Calculated Image	Off

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	epse
Dimension	2D
Excitation	Standard
Gradient Mode	Performance
Bandwidth	1842 Hz/Px
Echo Spacing	0.65 ms
Free Echo Spacing	Off
EPI Factor	118

Sequence - Part 2

Introduction	Off
RF Spoiling	Off
Phase Correction	Internal

Sequence - Special

Excite pulse duration	3200 us
Refocus pulse duration	7040 us
Min. prep scans	0
Min. prep scans SB	1
Delay before PC scans	0 us
Single-band images	Off
MB LeakBlock kernel	On
MB dual kernel	Off
MB RF phase scramble	Off
Opt. MB RF pulse BW	Off
Time-shifted MB RF	Off
SENSE1 coil combine	On
Invert RO/PE polarity	Off
Force Maxwell corr.	Off
PF omits higher k-space	Off
Disable B1 control loop	Off

Sequence - Special

Disable freq. update	Off
Suppress 16-bit DICOM	Off
Force equal slice timing	Off
Online multi-band recon.	Online
FFT scale factor	1.00
Physio recording	Off

Sequence - Assistant

SAR Assistant	Off
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